

6G Weekly Survey — Aug 10 - Aug 16, 2026

15 items across 3 topics, ranked by importance.

6G Demos & Trials

New Mid-Band (FR3, 6-24 GHz) XL-MIMO for 6G: Channel Modeling, Algorithm Evaluation, and Field Trials

Haiyang Miao, Jianhua Zhang, Feifei Gao, Pan Tang, et al. · arXiv eess.SP · 2026-08-04

★★★★★ This work combines spectrum planning (likely to inform 3GPP Release 20+), custom measurement infrastructure at unprecedented scale (1536 elements), rigorous channel modeling for multiple XL-MIMO variants, and real-world field trials—directly addressing key 6G standardization gaps in FR3.

This paper systematically reviews FR3 (6-24 GHz) spectrum for 6G and extremely large-scale MIMO (XL-MIMO) with 768-1536 elements, covering spectrum allocation, a custom wideband massive MIMO channel sounder, propagation characteristics including near-field effects and spatial non-stationarity, and AI-assisted signal processing. The authors evaluate four XL-MIMO architectures (co-located, cell-free, intelligent) through modeling and conduct field trials in Upper 6 GHz, demonstrating that target SNR critically affects practical XL-MIMO performance. Results provide empirical validation of new mid-band XL-MIMO feasibility and highlight deployment considerations for next-generation systems.

Contribution: First comprehensive integration of FR3 spectrum policy review, custom thousand-element channel sounder development, multi-architecture XL-MIMO channel modeling with near-field/non-stationary effects, and field trial validation in Upper 6 GHz band.

Relevance to your work: Directly relevant: provides validated channel models and measurement data essential for training AI-native PHY algorithms in massive MIMO scenarios, plus empirical benchmarks for beam management and CSI compression schemes you develop for beyond-5G systems.

Cross-Layer Optimization and System-Level Design of Next-Generation Wireless Networks via Intelligent RAN Control

Maria Tsampazi · arXiv cs.NI · 2026-08-03

★★★★■ Comprehensive integration of AI-native RAN control, RIS modeling, and NTN spectrum sharing with extensive experimental validation on industry-relevant platforms (Colosseum, 5G testbeds) addresses multiple 6G research pillars simultaneously.

This dissertation addresses cross-layer optimization in next-generation wireless networks by integrating Open RAN principles, AI-driven control loops, and reconfigurable intelligent surfaces (RIS). Core contributions include DRL-based network slicing/scheduling, the PandORA framework for deploying AI agents on Colosseum emulator, RIS channel modeling and resource allocation across spectrum bands, and spectrum sharing between cellular and non-terrestrial networks via power control. Validation spans simulation, hardware-in-the-loop emulation, and over-the-air 5G testbed experiments.

Contribution: PandORA framework enables automated design, training, and deployment of DRL-based Open RAN applications on experimental platforms, bridging algorithmic research and real-world testbed validation.

Relevance to your work: Directly relevant: combines AI-native air interface design, RIS channel modeling, NTN integration, and provides experimental frameworks (PandORA, Digital Twins) applicable to validating learned PHY and cross-layer AI solutions for beyond-5G systems.

5G ISAC-Based UAV Detection and 3-D Tracking Using Uplink Sounding Reference Signals on an End-to-End O-RAN Simulation Testbed

Arun K. Gurung, Satha K. Sathananthan, Shiva R. Pokhrel · arXiv cs.NI · 2026-08-06

★★★★■ Strong demonstration of ISAC feasibility on standard 5G waveforms with open-source O-RAN components, directly relevant to 3GPP Rel-19 ISAC work and commercial deployment pathways.

The authors implement an end-to-end 5G ISAC testbed using OpenAirInterface, FlexRIC, and Sionna RT to detect and 3D-track low-altitude UAVs by repurposing NR uplink sounding reference signals as passive radar waveforms, with no modifications to the 3GPP standard. A bistatic configuration initially leaves elevation unobservable, which they resolve by adding either a vertical antenna aperture or a second transmitter for range diversity, achieving 1.8 m altitude RMSE with an EKF tracker exposed as an O-RAN xApp. Detection performance is maintained under concurrent 10 Mbps uplink communication traffic, demonstrating real coexistence of sensing and communication.

Contribution: First live O-RAN demonstration of passive-radar-based UAV 3D tracking using standard NR uplink SRS and an xApp-based tracker, establishing altitude observability through both array geometry and transmitter diversity without requiring a dedicated sensing waveform.

Relevance to your work: Directly relevant: provides an end-to-end reference implementation for ISAC using standard PHY signals, corroborates ray-tracing channel models in Sionna RT, and demonstrates sensing-communication coexistence metrics critical for 6G ISAC system design.

RIS-Enabled Energy-Efficient ISAC for Vehicular Applications

Shengyu Yang, Yuhan Wang, Shuhao Zeng, Haobo Zhang, et al. · arXiv eess.SP · 2026-08-05

★★★★■ Solid experimental validation of RIS+ISAC integration for a relevant 6G use case, though the sub-6 GHz operation and incremental nature limit broader impact compared to mmWave or more fundamental ISAC breakthroughs. This paper presents a prototype RIS-based ISAC system for vehicle-to-infrastructure networks operating in sub-6 GHz, demonstrating that a 6.8W RIS array enables simultaneous vehicle tracking (0.11m average localization error) and data transmission (41.9% EVM reduction) with lower power consumption than conventional phased arrays. The system addresses energy efficiency challenges of large-scale antenna arrays needed for beamforming-enhanced V2I connectivity and sensing. Experimental validation shows practical feasibility for intelligent transportation applications.

Contribution: First experimental demonstration of RIS-enabled ISAC for vehicular networks, showing practical energy-efficient implementation with quantified sensing accuracy and communication quality improvements over non-RIS baseline.

Relevance to your work: Directly relevant as experimental ISAC demonstration with practical V2I application, providing real-world performance benchmarks (localization error, EVM metrics) for RIS-assisted joint sensing-communication systems that inform future ISAC research directions.

Self-Healing 6G Networks-in-Network for Resilient Wireless Communication

Daniel Lindenschmitt, Anton Schösser, Afnan Aghai, Philipp Schulz, et al. · arXiv cs.NI · 2026-08-05

★★★★■ Solid incremental work on network resilience and spectrum sharing for 6G, with a functional demonstrator, but lacks breakthrough algorithmic contributions or performance evaluation against baselines.

This demonstration presents a self-healing networks-in-network architecture for 6G that integrates continuous spectrum monitoring with dynamic spectrum management to autonomously detect and mitigate interference or jamming. A live system allows observers to introduce controlled disruptions and watch real-time anomaly detection, frequency reallocation, and network recovery. The work targets resilient operation in dynamic radio environments where multiple autonomous sub-networks share spectrum.

Contribution: The novelty is the tight integration of spectrum anomaly detection with dynamic spectrum management in a closed control loop, enabling autonomous frequency reallocation and recovery without manual intervention.

Relevance to your work: Tangential—focuses on network-level spectrum management and self-healing control loops rather than AI-native air interface design, physical-layer ML, or advanced waveform/sensing techniques.

FedVAR: Prototype-Aligned Federated Framework for Video Anomaly Recognition

Ghani Haider, Majid Kundroo, Boyun Eom, Dong Hwan Park, et al. · arXiv cs.LG · 2026-08-07

★★☆☆ Competent federated learning work for industrial video analytics, but fundamentally an application-layer computer vision problem with limited direct impact on physical-layer 6G wireless system design.

The paper addresses semantic misalignment in federated learning for video anomaly recognition (VAR) in Industrial IoT, where distributed edge clients develop divergent interpretations of normal versus abnormal events. FedVAR uses vision-language models with prototype-based alignment to create shared semantic anchors, enforcing consistent normality representations across clients while enabling fine-grained anomaly classification. Experiments on non-IID data partitions and novel anomaly classes show consistent improvements over federated baselines.

Contribution: First federated framework for fine-grained video anomaly recognition that explicitly addresses semantic misalignment through VLM-based prototype alignment, moving beyond binary anomaly detection.

Relevance to your work: Tangential—represents an edge intelligence use case that would consume wireless resources in IIoT/digital twin verticals, but does not advance AI-native air interfaces, channel modeling, or physical-layer research.

MAUPITI: On-Device Prototype-Based Learning on a Smart Infrared Sensor

Beatrice Alessandra Motetti, Tanguy Dugard du Villard, Matteo Risso, Alessio Burrello, et al. · arXiv cs.LG · 2026-08-07

★★☆☆ Competent embedded ML work but narrowly focused on gesture recognition with low-resolution IR sensors; limited direct relevance to wireless PHY or 6G system architecture.

This work presents a privacy-preserving smart IR sensor (16×16 thermal array) with on-device prototype-based learning for pose/gesture recognition under extreme constraints (<32kB memory, ~1.5mW power). They use a Nearest Class Mean classifier with an offline-trained CNN encoder and on-device streaming prototype updates to avoid backpropagation memory overhead. Experiments show comparable accuracy to conventional classifiers with <0.29% latency overhead for both inference and adaptation.

Contribution: Demonstrates practical on-device continual learning on ultra-constrained embedded sensors by decoupling feature extraction (offline CNN training) from adaptation (streaming prototype updates), eliminating replay buffers and backpropagation memory costs.

Relevance to your work: Tangential—the on-device learning principles could inform edge intelligence for ISAC sensors or semantic communication endpoints, but the core contribution addresses computer vision on IR arrays rather than wireless air interface design.

FedDOSE: Federated Learning Framework Decomposing Site Effects for Modeling Brain Dynamic Functional Connectivity

Deepank Girish, Yi Hao Chan, Yubin Zheng, Sukrit Gupta, et al. · arXiv eess.SP · 2026-08-07

★★☆☆ While methodologically competent for medical imaging, this work addresses fMRI brain connectivity analysis with no connection to wireless communications, physical layer design, or 6G systems.

This paper proposes FedDOSE, a federated learning framework for analyzing dynamic functional connectivity in multi-site fMRI datasets, explicitly decomposing site-specific statistical heterogeneity. The method uses Tucker decomposition to encode high-dimensional dynamic FC tensors and aligns class-specific prototypes across sites using optimal transport and Procrustes analysis. Validation on autism and ADHD detection tasks across three multi-site datasets demonstrates improved performance over existing federated learning approaches.

Contribution: Novel combination of tensor decomposition for dynamic functional connectivity with explicit site-effect decomposition in federated learning, using optimal transport to align learned representations across heterogeneous medical imaging sites.

Relevance to your work: tangential

A foundation-model approach to pediatric headache classification from rs-fMRI

Guilherme S. Imai Aldeia, Clara Moon, Julie Shulman, Navil Sethna, et al. · arXiv cs.LG · 2026-08-07

★☆☆☆☆ This is a medical imaging application with no connection to wireless communications, physical layer design, channel modeling, or any aspect of beyond-5G networks.

This paper applies NeuroSTORM, a foundation model for fMRI data, to classify pediatric headache conditions from resting-state brain scans. The approach achieves AUROC 0.82 for headache vs. healthy controls and 0.69 for multiclass headache subtype classification, outperforming traditional functional-connectivity matrix methods. Results demonstrate transfer learning from fMRI foundation models under limited-data regimes.

Contribution: First application of a pre-trained foundation model to pediatric headache classification, showing that learned latent fMRI representations transfer without requiring hand-crafted functional-connectivity features.

Relevance to your work: tangential

Dual-Node NVIDIA DGX Spark over Tailscale: A Remote-Access Testbed for Distributed LLM Training and Cyber-Threat-Intelligence Fine-Tuning

Vasanth Iyer · arXiv cs.LG · 2026-08-07

★☆☆☆☆ Proof-of-concept LLM training on desktop hardware with limited novelty; unrelated to wireless PHY, air interface design, or 6G system research despite appearing under a '6G Demos & Trials' tag.

Authors demonstrate distributed pretraining of a small language model (NanoChat, depth-20) across two NVIDIA DGX Spark nodes with GB10 Grace Blackwell SoCs connected via 200 Gb/s direct fiber, sustaining ~1,890 tokens/s over 653M tokens. They document PyTorch DDP/NCCL configuration, interface binding, NCCL timeout debugging, and present a cyber-threat-intelligence fine-tuning use case on CISA advisories. The work establishes feasibility for multi-node LLM training on desktop-class accelerators but does not claim scaling efficiency.

Contribution: Provides a detailed reproducibility runbook for distributed LLM training on compact DGX Spark hardware, including remote administration over Tailscale VPN and troubleshooting of NCCL timeouts in a two-node topology.

Relevance to your work: Tangential—distributed training infrastructure may inform federated learning or edge-AI workflows for 6G, but this paper focuses on compute-cluster configuration for NLP tasks, not wireless communications.

6G Physical Layer

Task-Oriented Candidate-Latent Feedback for Coarse-to-Fine Sensing in Distributed OFDM-ISAC Networks

Shiv Shankar, Radha Krishna Ganti, J Klutto Milleth · arXiv eess.SP · 2026-08-04

★★★★★ Tackles a critical architectural bottleneck for distributed ISAC with a practical neural solution validated on realistic ray-traced scenarios, directly relevant to 6G split-processing and task-oriented communication paradigms.

The paper addresses SE-SF feedback bottleneck in distributed OFDM-ISAC by proposing a learned coarse-to-fine pipeline: a convolutional scorer generates delay-Doppler proposals at the sensing entity, a learned encoder compresses K candidate tokens (position, azimuth-elevation patches, confidence) into $bKC+18K+16$ bits, and the sensing function performs cross-candidate refinement for joint delay-Doppler-azimuth-elevation estimation. Ray-traced urban scenarios demonstrate 96-99% detection at 107-806 bytes per CPI (compression ratios $1.2-9.2 \times 10^4$ over raw DDAE tensors), reducing interface requirements from multi-Gbit/s to sub-Mbit/s. Cross-scene generalization to an unseen campus environment achieves 98.79-99.50% detection without retraining, indicating robust learned representations.

Contribution: First task-oriented, learning-based feedback design for distributed OFDM-ISAC that achieves four-orders-of-magnitude compression while preserving multi-target discrimination under clutter, with demonstrated cross-scene generalization.

Relevance to your work: Directly intersects your ISAC and task-oriented communication interests; the learned candidate-token encoder and cross-scene transfer results offer concrete design patterns for

semantic sensing feedback in next-generation distributed PHY architectures.

Prior-Aided Iterative Channel Reconstruction with Optimized Frame Structure for DSE Mitigation in CP-OTFS-Based LEO Satellite Systems

Yiyan Cheng, Tiejun Lv, Yashuai Cao, Xuehan Wang, et al. · arXiv eess.SP · 2026-08-04

★★★★■ Addresses a critical practical limitation (DSE) in OTFS for LEO satellites with a principled solution including theoretical bounds, relevant as NTN becomes part of 6G standards and OTFS gains traction for high-mobility scenarios. This paper addresses Doppler squint effect (DSE) in LEO satellite OTFS systems, where frequency-dependent Doppler shifts degrade delay-Doppler domain channel sparsity. The authors propose a CP-OTFS frame structure that leverages received signal energy distribution as prior information, combined with a prior-aided iterative channel reconstruction (PAICR) algorithm to mitigate DSE-induced power leakage. The frame design additionally reduces PAPR and OOB while the algorithm includes convergence guarantees and CRLB benchmarks.

Contribution: Novel OTFS frame structure exploiting energy distribution as prior information for DSE-aware channel estimation in LEO satellite systems, coupled with an iterative reconstruction algorithm that explicitly handles frequency-dependent Doppler effects.

Relevance to your work: Directly relevant to your NTN and 6G waveform interests, particularly the intersection of OTFS modulation with LEO satellite channel modeling and estimation under extreme Doppler conditions.

RIS-Aided mmWave Localization Under Cross-Link Interference via Beam-Domain ML Fingerprinting

Md Tarek Hassan, Dmitry Zelenchuk, Muhammad Ali Babar Abbasi · arXiv eess.SP · 2026-08-07

★★★★■ Solid incremental work addressing practical RIS beam management for 6G with realistic interference modeling, though limited to simulation with a single RIS size and frequency.

Proposes beam-domain fingerprinting for RIS-aided mmWave UE localization at 28 GHz using received SNR across predefined RIS reflection states, avoiding explicit CSI estimation. Evaluates four ML regressors (KNN, SVR, RF, MLP) under clean and cross-link interference conditions with a 20×20 RIS. Achieves 0.37° azimuth and 4 cm range MAE with KNN in clean scenarios, degrading to 1.4° and 7.6 cm under interference, with angle estimation significantly more sensitive to interference than range.

Contribution: Demonstrates that beam-domain RIS fingerprinting for localization exhibits asymmetric degradation under interference—azimuth estimates suffer more than range—revealing how location information is encoded differently across the fingerprint structure.

Relevance to your work: Directly relevant to RIS-assisted beam management and AI-native air interfaces; the beam-domain fingerprinting approach and interference analysis inform ML-based PHY design for beyond-5G massive MIMO and RIS deployment strategies.

ISAC (sensing + communication)

Sub-Sampling for Positioning Privacy in ISAC: Deception by Aliasing via Sparse Arrays and Pilots

L. Yashvanth, Christos Masouros, Suraj Srivastava, Aditya K. Jagannatham, et al. · arXiv eess.SP · 2026-08-07

★★★★■ Addresses the critical but underexplored problem of sensing-layer security in ISAC systems with a novel aliasing-based approach that leverages fundamental signal processing principles without performance degradation—highly relevant as ISAC becomes a 6G pillar.

This paper proposes a privacy-preserving ISAC framework that uses spatial undersampling (sparse arrays) and frequency undersampling (sparse pilot allocation) to create controlled aliasing, causing unauthorized receivers to observe ghost targets at incorrect range-angle positions. The authors derive closed-form expressions for the range-angle MIMO ambiguity function under sub-sampling and establish conditions under

which aliasing-induced ambiguities translate to positioning deception. The approach maintains legitimate ISAC performance while naturally obfuscating sensing information from eavesdroppers without additional overhead.

Contribution: First framework to exploit joint spatial-frequency sub-sampling as a deception mechanism for sensing privacy in ISAC, providing closed-form aliasing analysis and demonstrating that controlled undersampling creates ghost targets for unauthorized receivers while preserving legitimate sensing performance.

Relevance to your work: Directly relevant to your ISAC research; this privacy mechanism could integrate with AI-based beam management and channel estimation frameworks you work on, particularly for sparse array processing and pilot design in sensing-communication coexistence scenarios.

DRL-Based Secure Transmission for Rotatable Antenna-Enabled Low-Altitude ISAC Systems

Chuan Liu, Hongyi Bian, Wei Gao, Qi Zhang, et al. · arXiv eess.SP · 2026-08-07

★ ★ ■ ■ ■ Competent DRL application to a niche scenario (low-altitude ISAC with rotatable elements), but narrow scope limited to secrecy rate optimization without hardware validation or broader ISAC performance metrics.

This paper addresses secure transmission in a low-altitude ISAC system using rotatable antennas (RAs) that adjust 3D beam pointing per element. A multi-agent proximal policy optimization algorithm (MAPPO-T) jointly optimizes transmit beamforming and transmit/receive RA pointing matrices to maximize minimum secrecy rate against UAV eavesdroppers. Simulations show RA arrays improve secrecy rate over fixed-orientation antennas and the proposed MAPPO-T outperforms standard MAPPO.

Contribution: First application of per-antenna rotatable arrays to physical-layer security in low-altitude ISAC, using multi-agent deep RL to jointly optimize beamforming and 3D antenna orientations.

Relevance to your work: Directly relevant as it combines ISAC with DRL-based beamforming optimization, though the rotatable antenna hardware assumption may be too specialized for broader AI-native air interface design.